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13.17 Factors affecting Photosynthesis

In 1860, **Sachs** gave three cardinal points theory explaining minimum, optimum and maximum factors that control photosynthesis. In 1905, **Blackman** put forth the importance of smallest factor. **Blackman's law of limiting factor** is actually a modified Law proposed by **Liebig's Law of minimum**. According to Blackman, *"When a process is conditioned as to its rapidity by a number of separate factors, the rate of the process is limited by the pace of the lowest factor"*. To conclude in an easy way *"at any given point of time the lowest factor among essentials will limit the rate of photosynthesis"*. For example, when even sufficient light intensity is available, photosynthesis may be low due to low CO_2 in the atmosphere. Here, CO_2 acts as a limiting factor. If CO_2 is increased in the atmosphere the rate of photosynthesis also increases. Further increase in photosynthesis is possible only if the available light intensity is also increased proportionately (Figure 13.23).

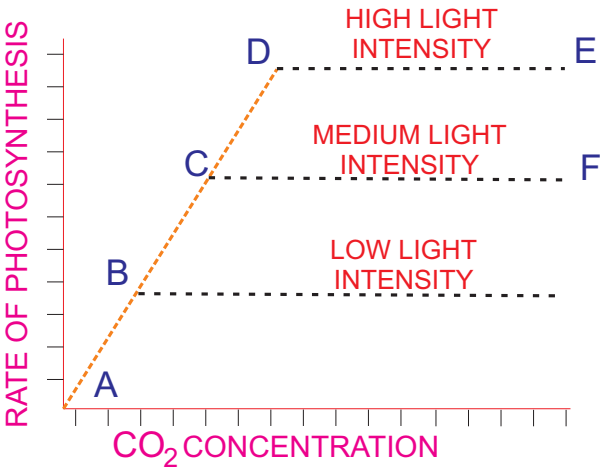


Figure 13.23: Blackman's Law of Limiting Factors

Factors affecting photosynthesis are further grouped into External or Environmental factors and Internal factors.

I. **External factors:** Light, carbon dioxide, temperature, water, mineral and pollutants.

II. **Internal factors:** Pigments, protoplasmic factor, accumulation of carbohydrates, anatomy of leaf and hormones.

13.17.1. External factors

1. Light

Energy for photosynthesis comes only from light. Photooxidation of water and excitation of pigment molecules are directly controlled by light. Stomatal movement leading to diffusion of CO_2 is indirectly controlled by light.

a. Intensity of Light:

Intensity of light plays a direct role in the rate of photosynthesis. Under low intensity the photosynthetic rate is low and at higher intensity photosynthetic rate is higher. It also depends on the nature of plants. Heliophytes (Bean Plant) require higher intensity than Sciophytes (*Oxalis*).

b. Quantity of Light:

In plants which are exposed to light for longer duration (Long day Plants) photosynthetic rate is higher.

c. Quality of light:

Different wavelengths of light affect the rate of photosynthesis because pigment system does not absorb all the rays equally. Photosynthetic rate is maximum in blue and red light. **Photosynthetically Active Radiation** (PAR) is between 400 to 700 nm. Red light induces highest rate of photosynthesis and green light induces lowest rate of photosynthesis.